Shell-Cheniere Global LNG Shock Relative-Value Model

Physical disruption -> regional gas prices -> company cash flow -> equity response

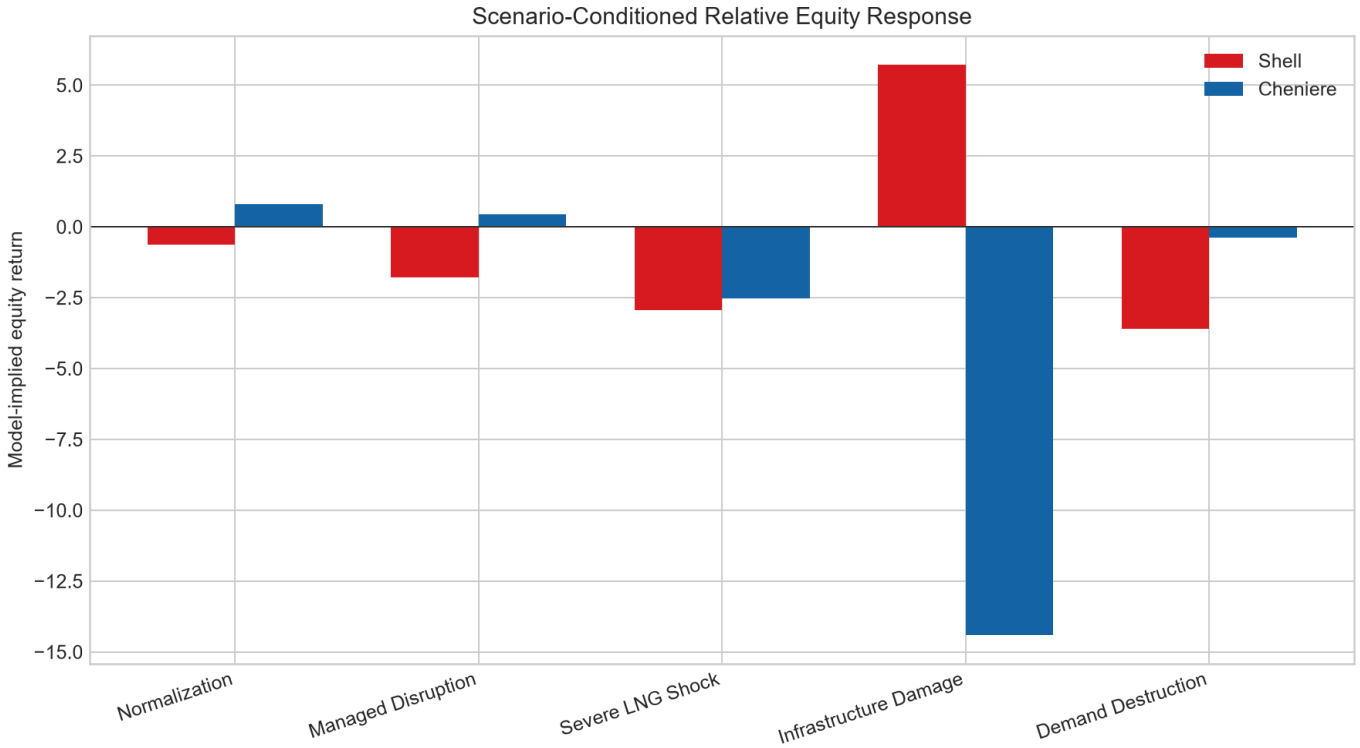
In the Severe LNG Shock calibration, the deterministic model modestly favors **Cheniere**. The Monte Carlo simulation assigns a **53.8%** probability that Cheniere outperforms Shell, indicating a narrow and assumption-sensitive relative edge.

Decision question

When Gulf LNG supplies are disrupted, which business model delivers stronger equity resilience: Shell's diversified LNG portfolio and trading optionality, or Cheniere's U.S.-anchored liquefaction platform?

Scenario results

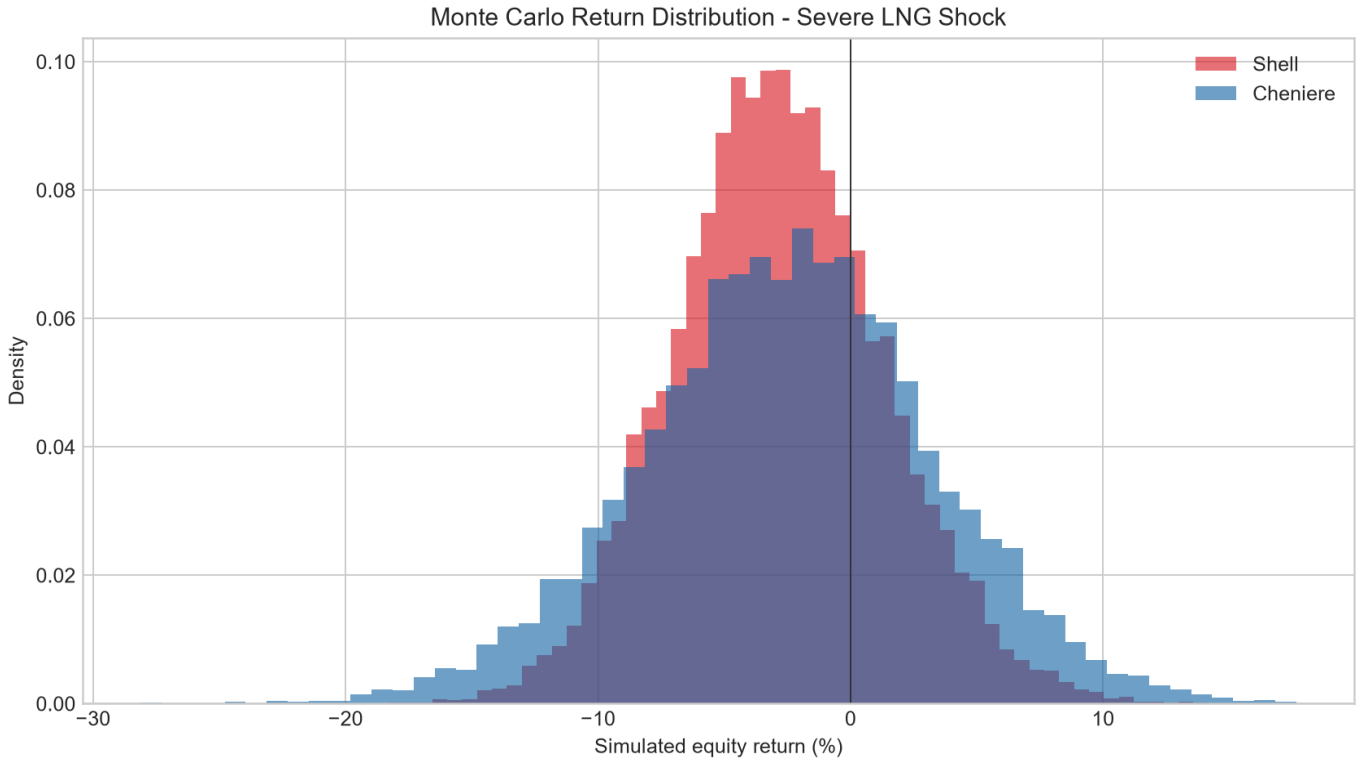
Scenario	Net gap	JKM shock	Shell	Cheniere	Preferred
Normalization	0.3%	3.0%	-0.6%	0.8%	Cheniere
Managed Disruption	3.1%	29.5%	-1.8%	0.4%	Cheniere
Severe LNG Shock	8.8%	94.8%	-2.9%	-2.5%	Cheniere
Infrastructure Damage	13.6%	215.5%	5.7%	-14.4%	Shell
Demand Destruction	4.0%	76.4%	-3.6%	-0.4%	Cheniere



Monte Carlo risk distribution

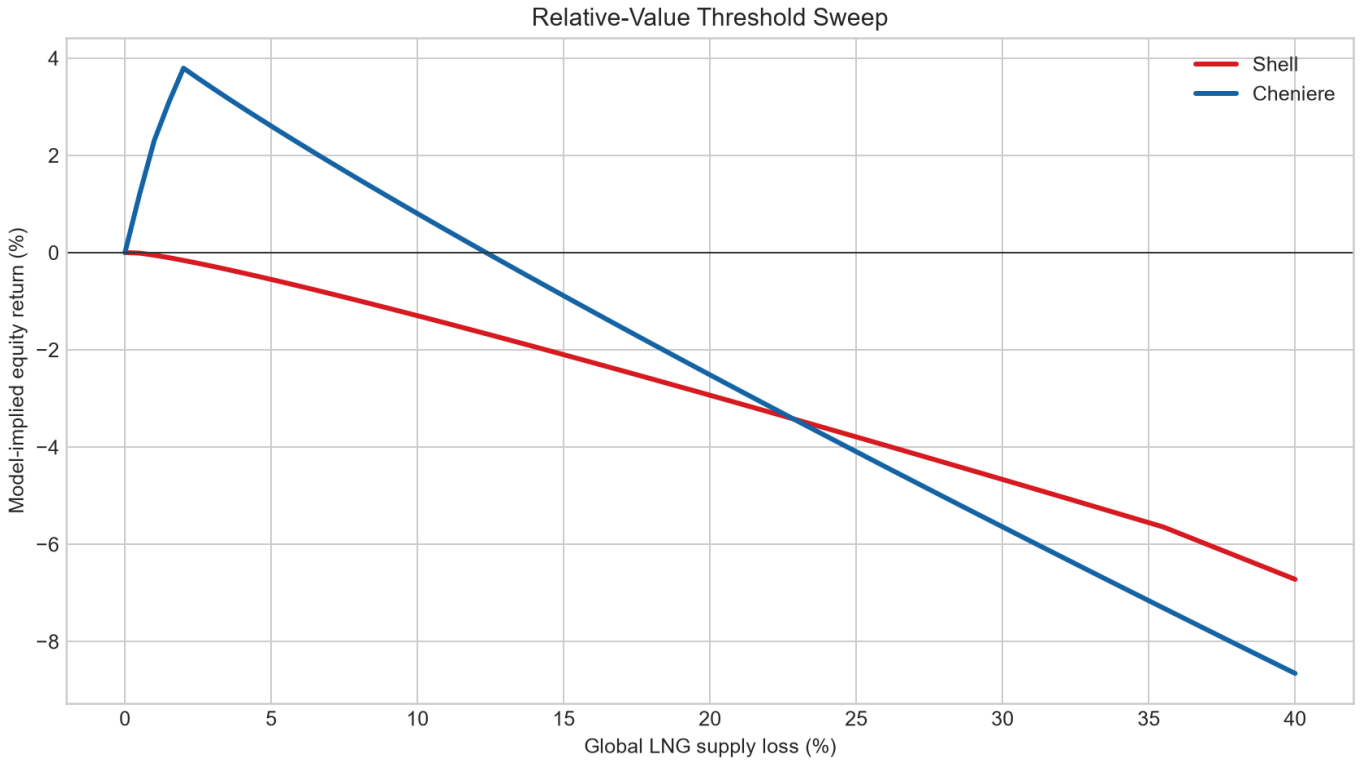
The model ran 10,000 simulations around the Severe LNG Shock scenario. Physical severity, duration, replacement supply, demand destruction, freight costs, price capture, and residual equity returns vary jointly.

Metric	Mean	P10	Median	P90	P(negative)
Shell equity return	-2.8%	-8.1%	-2.9%	2.6%	75.8%
Cheniere equity return	-2.5%	-9.9%	-2.4%	4.9%	66.8%
Cheniere minus Shell	0.4%	-7.0%	0.5%	7.4%	46.2%



Relative-value threshold

The first modelled preference crossover occurs near **22.8% global LNG supply loss**, with the remaining Severe LNG Shock assumptions held constant.



Transmission logic

Layer	Shell	Cheniere
Physical	Gulf-linked volumes; global portfolio	U.S. Gulf Coast capacity ceiling
Price	JKM/TTF exposure plus contract lags	International upside constrained by contracts
Costs	Freight, replacement cargoes, working capital	Henry Hub feed gas, utilization strain
Financial	Integrated Gas weighted within diversified group	Higher direct LNG cash-flow sensitivity

Methodology and governance

The model estimates conditional changes from a baseline; it does not forecast absolute share prices. Reported operational facts anchor the configuration. Non-public contract, trading, cargo, and price-capture exposures are explicit assumptions. The commodity engine converts the net supply gap into nonlinear JKM, TTF, and Henry Hub shocks. Company modules then apply volume, margin, freight, feed-gas, working-capital, utilization, and geopolitical valuation effects.

Calibration item	Value
Status	live
Observations	157
Shell market beta	0.14
Cheniere market beta	0.16
Shell annualized volatility	19.4%
Cheniere annualized volatility	28.3%
Return correlation	0.44

Core limitations

- Scenario outputs are sensitive to price elasticity and company price-capture assumptions.

- Shell trading and optimization economics are not publicly observable at cargo level.

- Cheniere contract structures reduce direct spot-price exposure and differ by agreement.

- The model does not include taxes, FX, derivatives, financing changes, or every segment interaction.

- Monte Carlo distributions describe model uncertainty, not objective investment probabilities.

Source register

ID	Source	Publisher	Use
SRC-001	About one-fifth of global LNG trade flows through Hormuz	U.S. Energy Information Administration	Calibrates 20 percent global LNG trade exposure to Hormuz
SRC-002	Gas Market Report Q3 2026	International Energy Agency	Frames Gulf supply loss and non-Gulf replacement dynamics
SRC-003	Shell Annual Report and Accounts 2025	Shell plc	Company reporting baseline and segment context
SRC-004	Shell Financial Modelling Guidance	Shell plc	Price-marker exposure and Integrated Gas modelling structure
SRC-005	Shell Q1 2026 results	Shell plc	Recent LNG sales and liquefaction-volume reference
SRC-006	Cheniere 2025 Form 10-K	U.S. Securities and Exchange Commission	Contract structure risk factors and financial baseline
SRC-007	Cheniere Q4 and FY 2025 results	Cheniere Energy	Calibrates operating liquefaction capacity
SRC-008	Cheniere Q1 2026 results	Cheniere Energy	Updates operating capacity to over 53 mtpa
SRC-009	Market prices via yfinance	Yahoo Finance	Optional equity beta volatility and correlation calibration

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